

In previous, class you have studied that clean water is a colourless, odourless and tasteless at room temperature. Water is also universal solvent because a wide range of chemical substances can easily be dissolved in it.

Whether it is the climate sustenance through water cycle or serving as the raw material for the green plants during photosynthesis or being one of the basic food component of humans and different animals, a constant supply of water is crucial for life.

Have you ever thought of how you get a constant supply of clean water? What are the sources of water? What are the uses of water? Earlier, you have also studied how water becomes polluted? How could this water be cleaned? How could we get clean water?

In this Chapter you will learn about:

- Water for Life
- Sources of Water
- Impurities of Water
- Cleaning of Water (Distillation, Water Treatment Plant)
- Uses of Water (Drinking, Source of Energy-Hydroelectricity, Cooling of Heavy Mechanical Complexes)

All the students will be able to:

- ✓ Describe the ways in which clean water is vital for meeting the needs of humans and other living things.
- ✓ Identify the sources of water.
- ✓ Recognise the substances present in water that make the water impure.
- ✓ Suggest different ways to clean the impure water.
- ✓ Describe the various uses of water in our country.
- ✓ Investigate the consumption of water in our daily life and suggest ways to reduce wastage of water.



Figure 5.1: Water for life

WATER FOR LIFE

DO YOU KNOW?

As 97.5% of water on earth is saltwater, therefore water recycling is regulated through seas and oceans.

- ✓ Describe the ways in which clean water is vital for meeting the needs of humans and other living things.

Water is essential for carrying out many of our living activities.

Clean water is colourless, odourless, germ-free and pleasant to taste. The clean water is considered safe for drinking. Water is required for other household, industrial, agricultural and environmental activities. Water is also used for recreational activities, such as in swimming pools, artificial ponds etc.

Can you imagine a life without water? No it is not possible. Water is essential for animals and plants. Clean water is also one of the six basic food components of human beings. Other five components are carbohydrates, proteins, fats, vitamins and minerals.

Green plants need water for photosynthesis. Can you imagine a life without green plants? It is not possible to survive without green plants, which are the direct or indirect source of food for animals and human beings.

As you have learned earlier, water is the only natural substance found in three states of matter as solid (in glaciers and icebergs), liquid (in form fresh water, rivers, lakes, streams and marine water seas and oceans) and gas (in water vapours, clouds and air). This unique physical property of water is important to maintain a constant supply of water to keep the environment healthy for the growth and evolution of living things.

Activity 5.1: List out the activities in which clean water is consumed by the humans.

What I need:

Notebook, pen/pencils, colour pencils.

What to do: (Duration 2 to 3 days)

Discuss the activities among your group members, observe your surroundings, talk to your elder siblings, parents and make a list of activities in which clean water can be used.

Activity 5.2: Complete the table below. Draw the use of water and display in classroom.

What I observed:

Household Activities	Agricultural Activities	Industrial Activities	Other Activities

Activity Questions:

Which household/agricultural/industrial/environmental activities can be performed by using clean water?

Water is one of the most crucial natural substances for the existence of any form of life on earth. Clean water is the basic requirement of human beings and most of the animals and plants. It is estimated that approximately 70% of freshwater is used for crop irrigation and livestock production. Industrial use is of 20% of the freshwater, while domestic use makes up the remaining 10 %.

Saltwater or marine (sea) water also contributes in important ways in the existence of life on earth. There are thousands of the animals and plant species that live in oceans and seas. These species play a crucial role in meeting the dietary needs of human beings and other animals. For instance, seaweeds are the healthy food source for human beings and also used in the making of fertilizers. Fishes such as Pomfret, Palla, Rahu and prawns are the most popular seafoods that are great source of protein, minerals like iodine and vitamins.

Teacher Note: Help students to explore. Display and appreciate their work.

In other words, human activities are directly influenced by the quality and availability of water. It is, thus very important to learn about water management.

SOURCES OF WATER

✓ Identify the sources of water.

Pakistan is blessed with both the natural and artificial sources of water. The main natural sources are rivers, lakes, streams, rain water oceans, glaciers and wells whereas artificial sources are water reservoirs, dams and tube wells.



Figure 5.2: Water Resources in Pakistan

In order to ensure the constant availability of clean water, the conservation of freshwater sources and preservation of physical properties of saltwater sources is important.

Activity 5.3: (a) Identify the sources of clean water in Sindh.

(b) Draw a labelled map of Sindh to show these sources with the names of districts and cities in which they are located.

What I need:

Different colour pencils and paper.

What to do:

1. Make a list of the sources of clean water in Sindh.
2. Draw the map of Sindh and mention the location of the clean water sources with the name of the associated districts, cities and rural areas.
3. Highlight the different sources with different colour pencils.

Activity Questions:

1. Where are the water sources mostly located?
2. State the population, livelihood of the people, the flora(plants/vegetation) and fauna (animals/livestock) of the district and cities with fresh water sources.

IMPURITIES OF WATER

✓ Recognise the substances present in water that make the water impure.

As you have studied earlier that water gets impure by the addition of untreated sewage water from domestic use, untreated water that contains harmful chemicals from our industries and water run-off from our agriculture that contains harmful pesticides and fertilizers. These impurities in water have severe impact on the health of Pakistani citizens, plants and animals.

These impurities include undissolved and dissolved substances such as sand, clay, peat, animal mass, decomposed vegetation, crop cuttings, microorganisms that cause diseases, chemicals from industries and domestic use, metal traces, Dichloro-diphenyl, Trichloro ethane (DDT) and agricultural run-off.

These impurities in water cause waterborne diseases such as hepatitis A, B and C, amoebic and other dysenteries, diarrhoea, typhoid and naegleria infection. Similarly these impurities cause increased rate of viral infected vegetables and fruits and different fungal infections in cattle in Sindh.

DO YOU KNOW?

Improved water is a term used to define the water free of any contamination while clean water is the term used to define water free of all physical and radiological impurities, and must have chemical and biological impurities in check [as per World Health Organisation (WHO) guidelines]. Only, then clean water is safe for drinking.

Worryingly, according to the Pakistan Water Situational Analysis Report (2014), Pakistan's freshwater sources are not safe for irrigation, household and industrial activities. Although, since 1990 access to improved water has been raised from 85% to 92%, still 64% rural population does not have access to drinkable water.

Water is termed as polluted water when it has any or all undesired and harmful physical, chemical and biological impurities. These impurities include disease causing microbes, clays, peat (decomposed vegetation, animal dung) sand particles and chemicals, such as metals and DDT.

CLEANING OF WATER

✓ Suggest different ways to clean the impure water.

Cleaning of water is essential to meet the vital needs of human beings and other living things. Is your drinking water improved or clean water?

You have studied earlier that the impurities remain either dissolved or undissolved in water. You have also studied that a variety of water cleaning methods used at domestic and commercial levels. The most commonly used water cleaning methods are boiling of water, filtration, distillation, chlorination, use of water purification tablets and potash alum. The use of portable and fixating water purifiers is common in the urban population.

Domestic Methods of Cleaning of Water:

Boiling: In villages and other less-privileged areas, boiling can easily be used to make water drinkable. Water is boiled for 15-20 minutes to kill disease causing microbes.

Potash Alum: It is used to clean water by sticking physical impurities together. As the impurities stick together to form large particles, these large enough particles settle down and are filtered.

Water Purifiers: Water purifiers are multi-effect water treatment systems, use filtration, Reverse Osmosis (RO), ionization and Ultra Violet (UV) rays to purify water for drinking and medicinal uses. Two examples of portable water purifiers are shown in figure 5.3 (a) and (b).



Figure 5.3 (a): Water filter using 4 stages-filtration



Figure 5.3 (b): Water purifier, using Ultra Violet (UV), RO and biocide (Agent to kill pathogenic microorganisms)

Water Purification Tablets:

Water purification tablets such as NaDCC (Sodium Dichloro isocyanurate) or Halazone tablets are also used to clean water by sticking physical impurities together. These tablets deactivate bacteria and viruses to make water drinkable.

Commercial Methods of Cleaning of Water:

Filtration: Filtration is the most primitive and effective method of removing suspended sand particles and insoluble physical impurities from water. These particles and impurities are removed by using filter paper as shown in figure 5.3 (c). However, many modifications in simple filtration have increased its efficiency. One such example is the addition of bio-film in the traditional yet effective filtration system that removes biological impurities as shown in figure 5.3 (d).

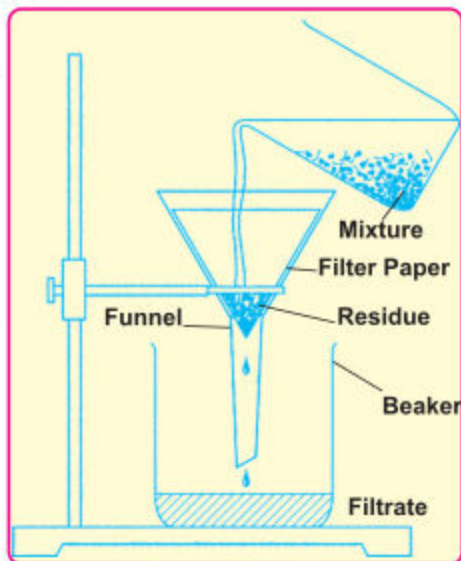


Figure 5.3 (c): Filtration by filter paper

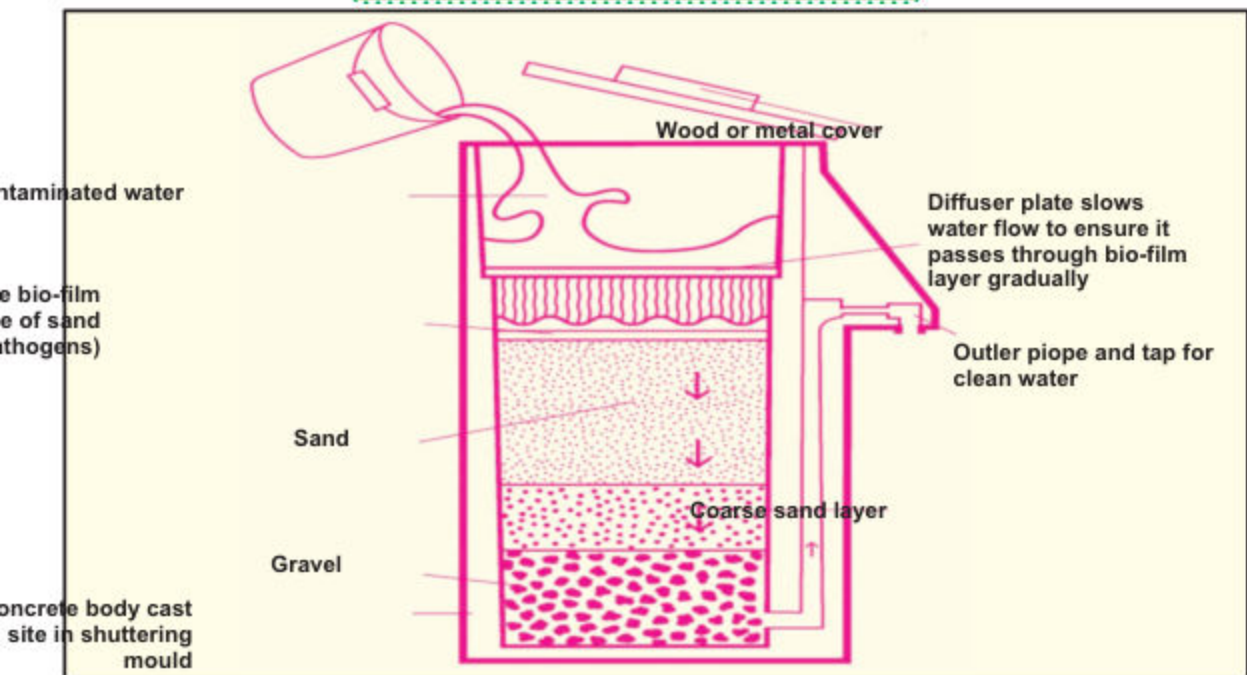


Figure 5.3 (d): Multi-stage filtration with a low-cost bio-film-A

Chlorination: Chlorination is another way to kill harmful microorganisms. It prevents the spread of water born diseases such as cholera, dysentery, typhoid etc. Liquid chlorine is mixed in the impure water to kill disease causing microorganisms. Use of chlorine oxide (ClO_2) and hypochlorite ions (ClO^-) makes water safe for the domestic use.

Distillation: Distillation is used to remove dissolved physical and chemical impurities from water. These impurities are removed by evaporating and condensing the polluted or impure water as shown in figure 5.3 (e). Distilled water is used for sterilizers as well as for preparing food for the patients. In laboratories, it is used in the preparation of suspensions and other medicines. However, distilled water is not advisable for the drinking. Can you find the reason, why is distilled water not advisable for the drinking.

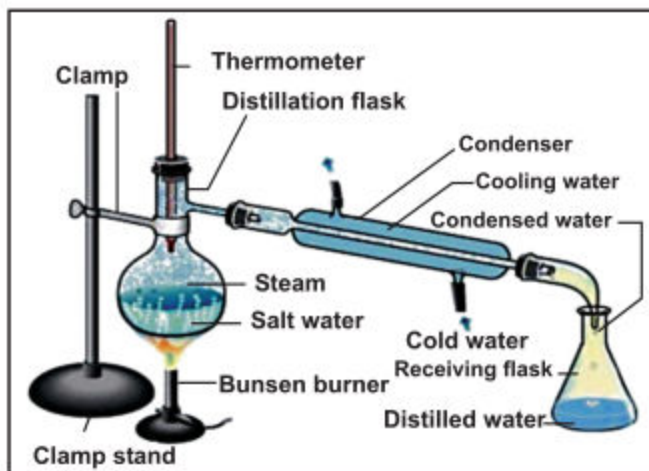


Figure 5.3 (e): Distillation in laboratory

Activity 5.4: Demonstrate cleaning of water through distillation

What I need:

1. One steel pot (diameter 12 inches) with compressed lid
2. A glass bowl (diameter 6-8 inches)
3. Salt solution (dissolve some salt into 3 litre of tap water) or salt water (3 litre)
4. Ice cubes
5. Stove and match box or lighter

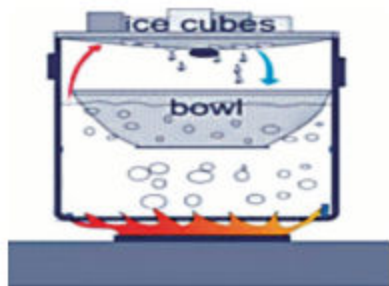


Figure 5.4 Collection of distilled water

What to do:

- Fill the pot with 3 litre of salt solution/salt water.
- Place the pot on the stove.
- Put the glass bowl in the pot (it should be floating on water).
- Cover the pot with lid (as shown in figure 5.4).
- Turn the stove on and let the water boil. When the sound of the popping water bubbles is heard, put ice cubes on the lid as shown in figure 5.4.

What I observed:

The steam will rise and strike to the inner part of the lid. As the lid must be cold due to the ice cubes, therefore steam will get condensed into pure water and fall in the bowl as shown in figure 5.4.

Activity Questions:

- Is distillation an adaptable method of water cleaning at small scale?
- How does distillation work to clean/purify water?

What I concluded:

Water cleaning plant:

In Pakistan to ensure supply of safe water to the rural and urban population several water cleaning plants are placed at different locations. These plants use multiple methods of water cleaning to remove physical, chemical and biological impurities from the water.

In Sindh, three different water qualities are maintained to serve the following purposes:

- (i) Improved water is used for the irrigation and commercial purposes including cooling of minerals and watering of sports grounds and public parks.
- (ii) Clean water is used for drinking, other household and pharmaceutical activities.
- (iii) Desalinated Arabian Sea water is used for household, industrial and recreational activities.

There are two ways of cleaning water.

1. Reverse Osmosis Filtration Plant:

Recently, the government of Sindh has installed a large number of the solar-powered reverse osmosis water filtration plants. These plants use reverse osmosis filtration method to remove physical, chemical and biological impurities and softening the underground water. Huge water tanks are being built near villages to make the clean water available for the drinking as well as other household activities.

DO YOU KNOW?

Osmosis is a process by which a solvent passes through a porous membrane.

Reverse Osmosis (RO) is a water purification technology that uses a semipermeable membrane to remove ions, molecules, and larger particles from drinking water.



Figure 5.5: Solar-powered RO filtration plant

2. Water Treatment Plant

Water treatment plant (as shown in figure 5.6) use multiple methods to remove hazardous chemicals, microorganisms, suspended particles of soil, peat, pesticides and many such contaminants. At first freshwater from a lake or reservoir is introduced to flocculation plant where flocculates are added to this water. These flocculates cause dust and pesticides to stick together.

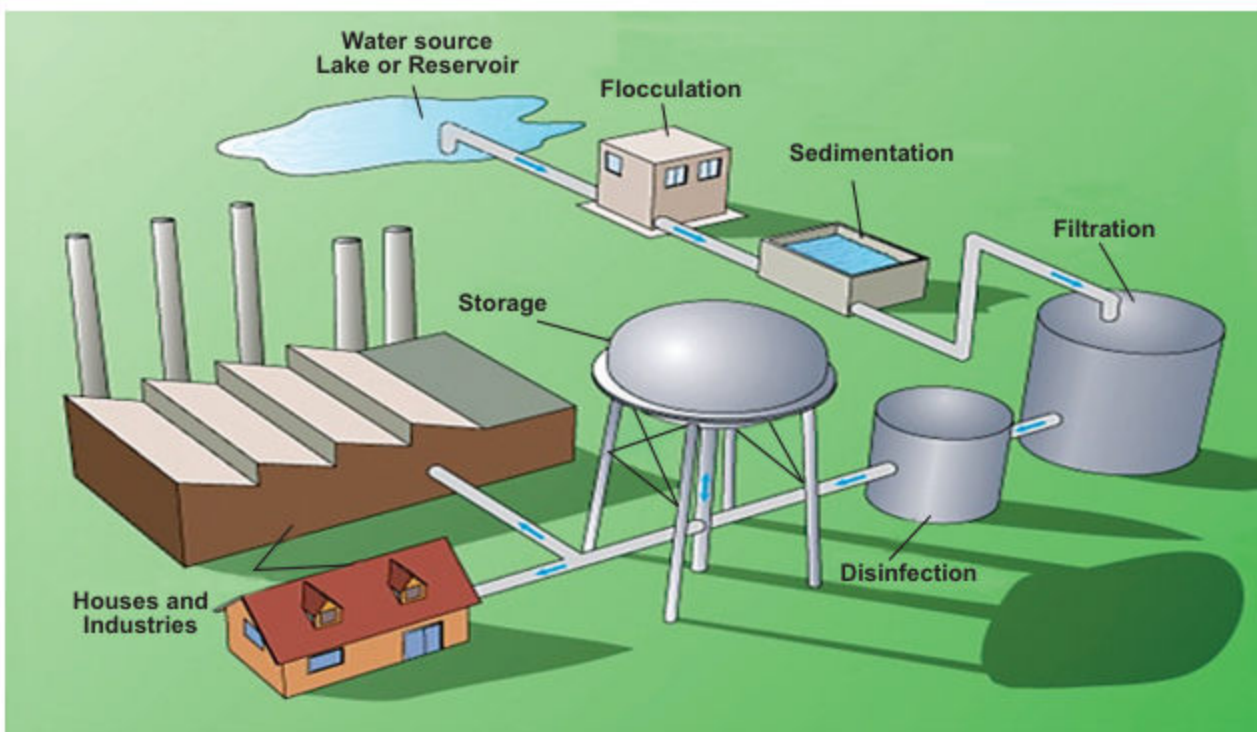


Figure 5.6: Drinking water treatment plant

Then, this water is transferred to the sedimentation tank and allows water to sit for 24 hours. This process helps in settling the particles down in the tank and clears the turbidity of freshwater along with flocculated pesticides. Now water is introduced into filter plant that has layers of charcoal, gravel and sand. When water seeps through these filters, all small particles are removed. At the last stage, water is transferred to the disinfectant tank to kill pathogenic microorganisms. Sometimes, minerals such as fluoride and sodium salts are added to the water to make it healthy and palatable.

USES OF WATER

✓ Describe the various uses of water in our country.

In Pakistan, usage of water can be divided into four broad categories:

(i) **Agricultural Use:** Approximately 93% of water is used in agricultural activities including irrigation of land, cattle farming and production of dairy food.

(ii) **Industrial or Commercial Use:** Approximately 5% of water is used in industry including food, pharmaceutical, textile, chemical industries and watering of public parks, sports complexes, wetland and amusement parks.

Water is the most efficient and economical coolant, used in heavy mechanical industries and nuclear reactors.

(iii) **Domestic Use:** Approximately 2% of water is available for the domestic use including drinking, cooking, cleaning, washing, bathing and other household activities.

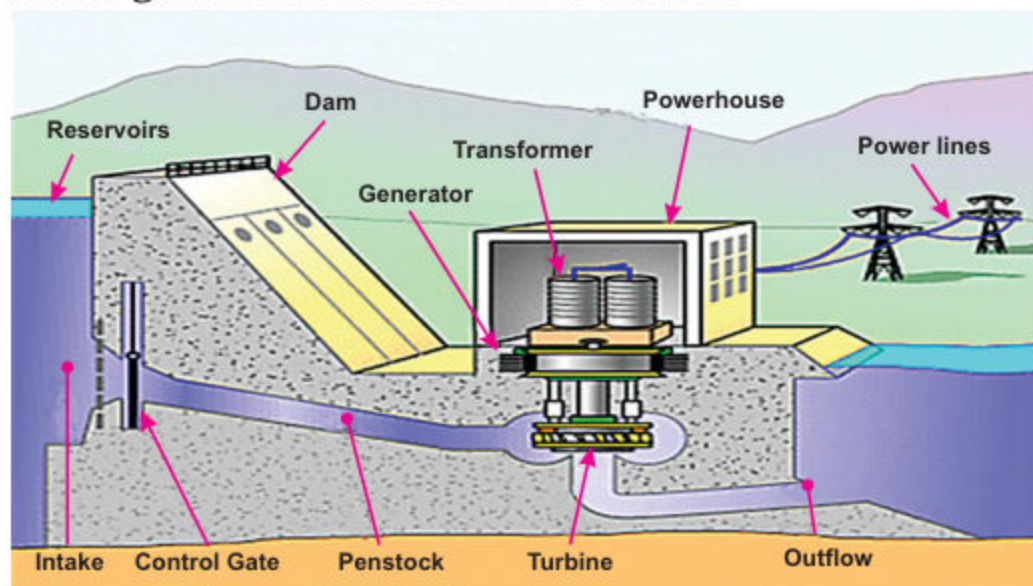


Figure 5.7 Hydroelectric power production

(iv) **Hydroelectric Use:** Water stored in reservoirs (such as dams) is used for the electric power production as shown in figure 5.7.

DO YOU KNOW?

Approximately 6913 Mega Watt of electricity is produced by different hydro power plants located at different regions of Pakistan.

CONSUMPTION OF WATER IN DAILY HUMAN LIFE

- ✓ Investigate the consumption of water in the daily human life and suggest ways to reduce wastage of water.

We use about 12% of water for drinking and cooking, 25% in washing, whereas 56% is used in bathrooms and toilets. About 2% water is used for plants in home but 5% is wasted through leakage of pipes. Figure 5.8 shows consumption of water in daily human life.

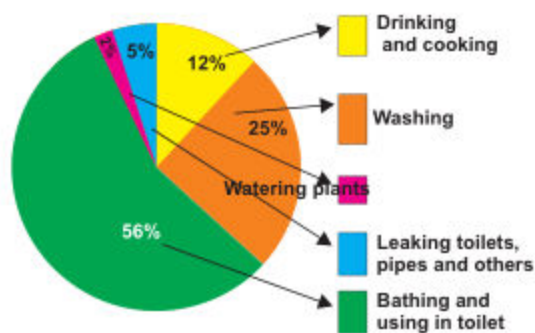


Figure 5.8: Water consumption in everyday life

Activity 5.5: Investigate the consumption of clean water in our daily life and suggest the ways to reduce wastage of water.

What I need:

Notebook, pen/pencil.

What to do:

Observe and enlist the daily activities in which clean water is consumed. Here, daily activities include household activities as well as the activities at school.

Now, identify the daily activities in which you and your family waste water.

Think a different techniques to reduce this daily wastage of water. Make a list of the most effective and easily adaptable ways to reduce the wastage of water.

What I observed:

S.No.	Activities in which water is consumed on daily basis	Ways,water is wasted in the daily consumption	Ways to reduce the wastage of water

Activity Questions:

1. Name the ways and activities for which you consume water at the most on the daily basis.
2. How is water wasted at most in your daily consumption?
3. How can the wastage of water be reduced at your home and school?

Ways to Reduce Wastage of Water:

Water scarcity is a global issue. Unfortunately, Pakistan is one of the top 20 countries under fierce threat of water scarcity. We need to be more vigilant in managing our water resources. Following are some effective ways to reduce wastage of water in our daily lives:

- (i) Make sure, there is no leakage of water storage tanks, utensils and pipe lines.
- (ii) Do not leave the water running while doing ablution or washing face and hands.
- (iii) Use water carefully in bathing. If you have a shower, then take short showers. Do not leave the shower on while applying soap on your body.
- (iv) Do not leave the water running while you are washing the dishes or clothes or cleaning vegetables and fruits. It would be wise to take water in a bucket to clean vegetables and fruits.
- (v) If using the washing machine, use it for only full loads.
- (vi) Water your lawns or potted plants in early hours or after 3:00 O'clock in the afternoon. It will help to restore their moisture for long.
- (vii) Use drought-resistant plants and shrubs for your lawns and home plantation.

Teacher Note: Ask each student to investigate the ways and activities for which she/he and their families consume water in their daily life.

Summary

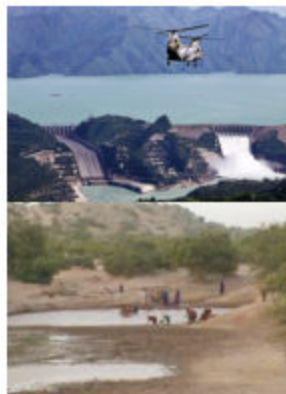
Water (H₂O)

(The most abundant natural source, essential for life)

Crucial for keeping environment healthy, used by green plants for photosynthesis, one of the basic food components of humans and animals.

Freshwater (3% of total Surface Water)

Saltwater (97% of total Surface Water)



Water Resources in Pakistan

Impurities in Pakistani Water Sources

Clay, peat, sand particles, disease causing pesticide, chemicals such as traces of metals and DDT.

Cleaning of Water

Domestic Level

Boiling of water, using potash alum or water purification tablets and water purifiers.

Commercial Level

Chlorination, filtration and distillation and water cleaning plants such as Tharparkar's RO filtration plant.

Uses of Water

Used in household, agricultural, industrial, recreational and environmental activities such as drinking, cooking, bathing, washing, cultivation of crops, watering of decorative plants, parks and sports grounds/ complexes, cooling of heavy mechanical complexes and nuclear reactors, pharmaceutical industry, hospitals, and in swimming pools and water-parks.

Review Exercises

1. Fill in the blanks:

- The process of water seepage in the land is called_____.
- In water cleaning plants, sand particles and pesticides are stuck together by using_____.
- Water is boiled for _____minutes to use it for drinking purpose.
- Seaweed is one of the important source of proteins, minerals of _____and vitamins.

2. Match the statements of column A with column B:

Column A	Column B
Potash Alum	Is used to remove dissolved physical and chemical substances.
Liquid Chlorine	Is used to kill harmful microorganisms.
Distillation	Is used to clean water by sticking physical impurities together to form large particles.

3. Explain the factors responsible for:

- Depleting freshwater sources in Pakistan.
- Deteriorating water-quality of clean water sources in Pakistan.

4. Describe the ways in which water is vital for the life of animals and plants.

5. Explain, Why:

- Boiled water is safe for drinking and cooking.
- Saltwater is desalinated.
- Clean water is important for sustainable healthy climate.

6. Suggest five ways to reduce the wastage of water in daily life.

7. Explain the processes of cleaning water in water treatment plants.

PROJECT

Use the following materials and design a model of water purification plant:

RO filters/semi permeable membranes, transparent water jars pipes for connecting different apparatus (having tap water and for storing clean water); flocculants such as Alum.

- a) Make a labelled diagram of your model of water cleaning system.
- b) Make a list of physical, chemical and biological impurities which can be removed by your designed water cleaning plant.
- c) Describe how your model works.
- d) Make a poster for advertisement of your water purification model plan in the school.